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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Visitor Entry Logging and Security Monitoring System

Submitted by

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DECLARATION

I hereby declare that the project entitled "Visitor Entry Logging and Security Monitoring System" submitted in partial fulfillment of the requirements for the course Java Programming is a **record of my original work** carried out under the guidance of **Mr.Aravind**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Pratham.S

CERTIFICATE

This is to certify that the project entitled **"**Visitor Entry Logging and Security Monitoring System**"** is a **bona fide work** carried out by **Pratham.S(2025-I-B.Sc.CS-'B') KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date: 10-04-2026

Submitted for the Viva-Voce Examination held on 10-04-2026

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ABSTRACT

The Visitor Entry Logging and Security Monitoring System is a Java-based application designed to manage and monitor visitor entries in organizations, offices, colleges, and residential buildings. In many places, visitor details are recorded manually in registers, which may lead to missing records, security risks, and difficulty in tracking visitor information. This project aims to automate the visitor entry process by providing a digital system that records visitor details and maintains a secure log.The system allows security personnel to enter visitor information such as visitor name, mobile number, purpose of visit, and automatically records the date and time of entry. The application validates mobile numbers, stores visitor records, and displays them in a structured format. The system also includes a search feature to quickly find visitor information when required.The project is developed using Core Java programming concepts such as classes, objects, methods, loops, conditional statements, and exception handling. The application is implemented using Java Swing to provide a graphical user interface (GUI) for easy interaction. The main objective of the system is to improve security monitoring, maintain proper visitor records, and reduce manual work. This project demonstrates the practical implementation of Java programming in developing a real-world security management application.

1. INTRODUCTION

Security is an important aspect in offices, colleges, apartments, and other organizations. Traditionally, visitor entries are recorded manually in notebooks or registers at the security desk. This manual process is time-consuming and may lead to incomplete or incorrect information. It also becomes difficult to track visitors during emergencies or verify previous records.

With the advancement of technology, automated visitor entry systems have become necessary to maintain proper records and improve security monitoring. A computerized visitor logging system allows security personnel to quickly enter visitor details and maintain digital records.

The Visitor Entry Logging and Security Monitoring System is a Java-based application designed to automate visitor entry management. The system records visitor name, mobile number, purpose of visit, and automatically captures date and time. It displays visitor records in a table format and provides search functionality.

This project is developed using Core Java and Java Swing GUI. The main goal of the system is to improve security, reduce manual errors, and provide quick access to visitor records.

1.1 Objective

The main objective of the Visitor Entry Logging and Security Monitoring System is to develop a Java-based application that automates visitor entry management.

Specific objectives include:
• To develop a Java GUI-based visitor entry system
• To record visitor details digitally
• To capture date and time automatically
• To validate mobile number entries
• To maintain visitor records securely
• To provide search functionality for quick access
• To improve security monitoring

1.2 Overview

The Visitor Entry Logging and Security Monitoring System is a GUI-based application developed using Java Swing. The system provides a digital platform where security personnel can enter visitor details and store them automatically.

The user enters visitor name, mobile number, and purpose of visit. The system captures the current date and time automatically and stores the record. All visitor entries are displayed in a table. The system also includes a search option to find visitors quickly.

This application reduces manual record keeping and improves security monitoring.

1.3 Features

The system includes the following features:

- Visitor Entry Form
- Mobile Number Validation
- Automatic Date & Time Capture
- Visitor Record Table Display
- Search Functionality
- Login Authentication
- User-Friendly Interface
- Fast Data Entry
- Security Monitoring Log

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the system requirements are studied in detail. It helps in understanding the problem, identifying the limitations of the existing system, and designing an effective solution. This phase ensures that the developed system meets user needs and functions efficiently.

2.1 Problem Definition

In many organizations such as offices, colleges, apartments, and hospitals, visitor details are recorded manually in registers. This traditional method has several drawbacks:

• Manual entry can lead to errors in recording visitor details
• Time-consuming process, especially during peak hours
• Difficulty in searching previous records
• Risk of losing or damaging physical records
• Lack of proper monitoring and security tracking
• No instant access to visitor information

These issues reduce efficiency and may create security risks. Therefore, there is a need for a computerized system that can store, manage, and retrieve visitor data accurately.

2.2 Project Overview

The Visitor Entry Logging and Security Monitoring System is a Java-based GUI application that records visitor information digitally. The system allows security personnel to:

• Enter visitor details
• Validate mobile number
• Capture date & time automatically
• Store visitor records
• Display records in table format
• Search visitor details

2.3 Scopes and Limitations

Scope

The system can be used in:
• Offices
• Colleges
• Apartments
• Hotels
• Companies
• Hospitals

Limitations

• No database storage (temporary memory only)
• No photo capture feature
• No ID card scanning
• Single-user system

3. SYSTEM DESIGN

System design is the phase where the overall structure of the system is planned before implementation. It defines how the system will work, how different components interact, and how data flows within the system. A well-designed system improves efficiency, maintainability, and usability.

The Visitor Entry Logging and Security Monitoring System is designed as a **Java Swing GUI-based application**. It provides a simple interface for users to enter visitor details, store records, and monitor visitor activity.

3.1 Modules

1. Login Module

This module provides authentication for security personnel.
Username and password are required to access the system.

2. Visitor Entry Module

This module allows entering:
• Visitor Name
• Mobile Number
• Purpose of Visit
• Date & Time (Auto)

3. Validation Module

This module validates:
• Mobile number must be 10 digits
• Fields should not be empty

4. Record Display Module

Displays visitor entries in table format including:
• Name
• Mobile Number
• Purpose
• Date & Time

5. Search Module

Allows searching visitor records by:
• Name
• Mobile number

6. Exit Module

Allows user to close the system safely

4. METHODOLOGY

The development process includes:

• Problem Analysis
• System Design
• Program Development
• Testing
• Documentation

4.1. Problem Analysis

Manual visitor logging is inefficient and insecure. A digital system is required to maintain records and improve monitoring.

4.2. System Design

System is divided into modules:
• Login Module
• Visitor Entry Module
• Validation Module
• Record Display Module
• Search Module

4.3. Program Development

The system is implemented using Java Swing GUI.

Core Java concepts used:
• Classes
• Objects
• Methods
• Loops
• Conditional Statements
• Exception Handling

4.4. Testing and Validation

Testing and validation are important stages in the development of the Visitor Entry Logging and Security Monitoring System to ensure that the application works correctly and meets the required objectives. The system was tested using different types of inputs, including valid and invalid data, to verify its functionality. Various test cases were applied, such as entering correct visitor details, providing invalid mobile numbers, leaving input fields empty, and performing search operations. The system successfully validated mobile numbers by allowing only 10-digit inputs and displayed appropriate error messages for incorrect entries. It was also verified that the date and time were captured accurately and that visitor records were stored and displayed properly in the table. The search functionality was tested to ensure quick and accurate retrieval of records. Overall, the system performed efficiently without errors, producing accurate results and confirming that it meets the intended requirements.

4.5. Result and Documentation

The system successfully records visitor details and displays them in table format. The search function helps retrieve records quickly.

4.6. Algorithm

1. Start program
2. Show login page
3. Enter username and password
4. Display visitor entry form
5. Enter visitor details
6. Validate mobile number
7. Capture date & time
8. Store record
9. Display in table
10. Search if required
11. Exit program

4.7. Flowchart

Start
↓
Login
↓
Enter Visitor Details
↓
Validate Mobile
↓
Save Entry
↓
Display Table
↓
Search (Optional)
↓
End

5. SOURCE CODE

```
import javax.swing.*;
import javax.swing.table.DefaultTableModel;
import java.awt.*;
import java.awt.event.*;
import java.text.SimpleDateFormat;
import java.util.Date;

public class VisitorEntrySystem {

    JFrame loginFrame, mainFrame;
    JTextField usernameField;
    JPasswordField passwordField;

    JTextField nameField, mobileField, purposeField, searchField;
    JLabel timeLabel;
    DefaultTableModel model;
    JTable table;

    // Login Page
    public VisitorEntrySystem() {
        loginFrame = new JFrame("Visitor Entry Login");
        loginFrame.setSize(350, 220);
        loginFrame.setLayout(null);
        loginFrame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        JLabel userLabel = new JLabel("Username:");
        userLabel.setBounds(30, 30, 80, 25);
        loginFrame.add(userLabel);
```

```
    usernameField = new JTextField();
    usernameField.setBounds(120, 30, 150, 25);
    loginFrame.add(usernameField);

    JLabel passLabel = new JLabel("Password:");
    passLabel.setBounds(30, 70, 80, 25);
    loginFrame.add(passLabel);

    passwordField = new JPasswordField();
    passwordField.setBounds(120, 70, 150, 25);
    loginFrame.add(passwordField);

    JButton loginBtn = new JButton("Login");
    loginBtn.setBounds(120, 110, 100, 30);
    loginFrame.add(loginBtn);

    loginBtn.addActionListener(e -> login());

    loginFrame.setVisible(true);
}

void login() {
    String user = usernameField.getText();
    String pass = new String(passwordField.getPassword());

    if (user.equals("admin") && pass.equals("1234")) {
        loginFrame.dispose();
        showMainSystem();
    } else {
        JOptionPane.showMessageDialog(loginFrame, "Invalid Login");
```

```
    }
}

// Main System
void showMainSystem() {
    mainFrame = new JFrame("Visitor Entry Logging & Security Monitoring");
    mainFrame.setSize(800, 500);
    mainFrame.setLayout(null);
    mainFrame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

    JLabel title = new JLabel("Visitor Entry Form");
    title.setFont(new Font("Arial", Font.BOLD, 16));
    title.setBounds(20, 10, 250, 30);
    mainFrame.add(title);

    JLabel nameLabel = new JLabel("Visitor Name:");
    nameLabel.setBounds(20, 50, 100, 25);
    mainFrame.add(nameLabel);

    nameField = new JTextField();
    nameField.setBounds(130, 50, 150, 25);
    mainFrame.add(nameField);

    JLabel mobileLabel = new JLabel("Mobile No:");
    mobileLabel.setBounds(20, 90, 100, 25);
    mainFrame.add(mobileLabel);

    mobileField = new JTextField();
    mobileField.setBounds(130, 90, 150, 25);
    mainFrame.add(mobileField);
```

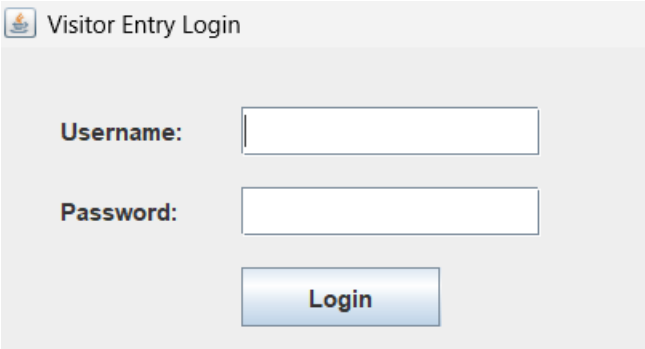
```
JLabel purposeLabel = new JLabel("Purpose:");
purposeLabel.setBounds(20, 130, 100, 25);
mainFrame.add(purposeLabel);

purposeField = new JTextField();
purposeField.setBounds(130, 130, 150, 25);
mainFrame.add(purposeField);

JLabel timeText = new JLabel("Date & Time:");
timeText.setBounds(20, 170, 100, 25);
mainFrame.add(timeText);

timeLabel = new JLabel();
timeLabel.setBounds(130, 170, 200, 25);
mainFrame.add(timeLabel);

JButton addBtn = new JButton("Add Entry");
addBtn.setBounds(50, 210, 120, 30);
mainFrame.add(addBtn);

JButton clearBtn = new JButton("Clear");
clearBtn.setBounds(180, 210, 100, 30);
mainFrame.add(clearBtn);

// Table
model = new DefaultTableModel();
model.setColumnIdentifiers(new String[]{
        "Name", "Mobile", "Purpose", "Date & Time"
});

table = new JTable(model);
```

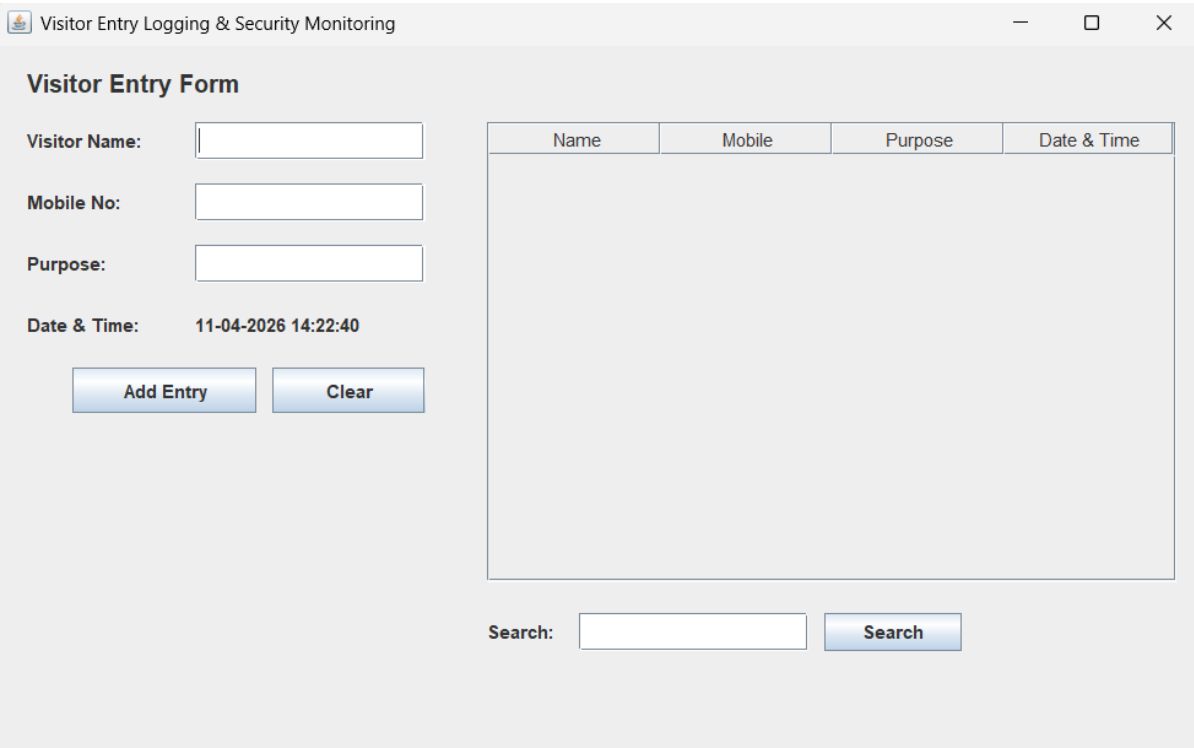
```
    JScrollPane pane = new JScrollPane(table);
    pane.setBounds(320, 50, 450, 300);
    mainFrame.add(pane);

    // Search
    JLabel searchLabel = new JLabel("Search:");
    searchLabel.setBounds(320, 370, 60, 25);
    mainFrame.add(searchLabel);

    searchField = new JTextField();
    searchField.setBounds(380, 370, 150, 25);
    mainFrame.add(searchField);

    JButton searchBtn = new JButton("Search");
    searchBtn.setBounds(540, 370, 90, 25);
    mainFrame.add(searchBtn);

    updateTime();

    addBtn.addActionListener(e -> addEntry());
    clearBtn.addActionListener(e -> clearFields());
    searchBtn.addActionListener(e -> searchEntry());

    mainFrame.setVisible(true);
}

void updateTime() {
    Timer timer = new Timer(1000, e -> {
        SimpleDateFormat sdf = new SimpleDateFormat("dd-MM-yyyy HH:mm:ss");
        timeLabel.setText(sdf.format(new Date()));
    });
```

```
    timer.start();
}

void addEntry() {
    String name = nameField.getText();
    String mobile = mobileField.getText();
    String purpose = purposeField.getText();
    String time = timeLabel.getText();

    if (!mobile.matches("\\d{10}")) {
        JOptionPane.showMessageDialog(mainFrame,
                "Mobile number must be 10 digits");
        return;
    }

    if (name.isEmpty() || purpose.isEmpty()) {
        JOptionPane.showMessageDialog(mainFrame,
                "Fill all fields");
        return;
    }
    model.addRow(new Object[]{name, mobile, purpose, time});
    clearFields();
}
void clearFields() {
    nameField.setText("");
    mobileField.setText("");
    purposeField.setText("");
}
void searchEntry() {
    String search = searchField.getText().toLowerCase();
```

```
        for (int i = 0; i < model.getRowCount(); i++) {
            String name = model.getValueAt(i, 0).toString().toLowerCase();
            String mobile = model.getValueAt(i, 1).toString();

            if (name.contains(search) || mobile.contains(search)) {
                table.setRowSelectionInterval(i, i);
                return;
            }
        }
        JOptionPane.showMessageDialog(mainFrame, "No Record Found");
    }
    public static void main(String[] args) {
        new VisitorEntrySystem();
    }
}
```

6. INPUT AND OUTPUT SCREENS





7. DISCUSSION & CONCLUSION

7.1 Discussion

The Visitor Entry Logging and Security Monitoring System automates visitor management. The system records visitor details, validates mobile numbers, and stores entries digitally. The search functionality helps retrieve information quickly. The system reduces manual work and improves security.

7.2 Conclusion

The Visitor Entry Logging and Security Monitoring System demonstrates the practical implementation of Java Swing GUI. It improves security monitoring and maintains visitor records efficiently. The system reduces manual errors and speeds up entry logging. The project can be enhanced by adding database connectivity, photo capture, and ID card scanning.

8. REFERENCES

Books

• Herbert Schildt – Java: The Complete Reference

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.javatpoint.com
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